Dong-Geun Kim

Ph.D. Candidate | POSTECH Interaction Lab

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Google Scholar · Publication data checked 2026-10-19

Research interests

Tactile communication · Vibrotactile perception · Mid-air haptics

Education

Pohang University of Science and Technology (POSTECH)

Integrated M.S./Ph.D. in IT Convergence Engineering

2020 - Present

Ulsan National Institute of Science and Technology (UNIST)

B.S. in Mechanical Engineering

2014 - 2019

Research experience

Research Assistant · POSTECH Kim Lab

Sep. 2019 - May 2020

Developed high-frequency ultrasound transducers for tactile applications.

Awards

2026 · Best Paper Award Finalist
IEEE Haptics Symposium

2024 · Best ToH Short Paper Award (Finalist)
IEEE Haptics Symposium

2022 · Best Paper Award
KSC (KIISE)

Publications indexed in Google Scholar

Titles and years follow the Google Scholar profile. Complete author names and existing PDF links are retained where available.

1. Effects of Spatiotemporal Parameters on Forearm Vibrotactile Stimulus Identification

Dong-Geun Kim, Geunho Lee, Suheon Nam, Chaeyong Park, Seungmoon Choi

2026 IEEE Haptics Symposium (HAPTICS), 1-13 · 2026

Best Paper Award Finalist

2. Investigating Luxuriousness of Steering Wheel Vibration Warning

K Kwon, SH Han, S Choi, B Jeong, J Kim, J Lee, DG Kim, W Lee

International Journal of Human-Computer Interaction, 1-21 · 2026

3. Tactile Enhancement of Mid-Air Ultrasonic Stimulation by Wrist Vibration: Perceived Intensity and Pattern Recognition

Dong-Geun Kim, Seungmoon Choi

IEEE Transactions on Haptics 18 (3), 603-614 · 2025

4. A feasibility study on tactile enhancement of mid-air ultrasonic stimulation by wrist vibration

Dong-Geun Kim and Seungmoon Choi

International Conference on Human Haptic Sensing and Touch Enabled Computer ... · 2024

5. Generating real-time, selective, and multimodal haptic effects from sound for gaming experience enhancement

Gyeore Yun, Minjae Mun, Jungeun Lee, Dong-Geun Kim, Hong Z Tan, and Seungmoon Choi

Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems ... · 2023

6. Sound-to-touch crossmodal pitch matching for short sounds

Dong-Geun Kim, Jungeun Lee, Gyeore Yun, Hong Z. Tan, and Seungmoon Choi

IEEE Transactions on Haptics 17 (1), 2-7 · 2023

7. MMGrip: A Handheld Multimodal Haptic Device Combining Vibration, Impact, and Shear for Realistic Expression of Contact

Dong-Geun Kim, Jungeun Lee, and Seungmoon Choi

SIGGRAPH Asia 2022 Posters, 1-2 · 2022

8. MMGrip: 사실적인 햅틱 표현을 위한 진동**,** 임팩트**,** 전단력 기반 다중촉감 햅틱 장치

Dong-Geun Kim, Jungeun Lee and Seungmoon Choi

한국정보과학회 학술발표논문집, 1038-1040 · 2022

9. Vibration-augmented buttons: Information transmission capacity and application to interaction design

Chaeyong Park, Jeongwoo Kim, Dong-Geun Kim, Seungjae Oh, and Seungmoon Choi

Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems ... · 2022

Additional presentations and posters

Preserved from the existing website; not part of the current Scholar publication list.

1. Tactile Localization in Forearm-Mounted Multi-Tactor Displays: The Effect of Tactor Number and Position

Dong-Geun Kim, Suheon Nam, Geunho Lee, Chaeyong Park and Seungmoon Choi

IEEE World Haptics Conference · 2025

2. A Feasibility Study of Tactile Enhancement of Mid-Air Ultrasonic Stimulation by Wrist Vibration

Dong-Geun Kim and Seungmoon Choi

Proceedings of Korea Haptics Conference · 2024

3. Investigating the Luxuriousness of Haptic Warning on the Steering Wheel

Jeongwoo Kim, Jungeun Lee, Dong-Geun Kim, Ki-Min Kwon, Beomyoung Chung, Junseong Park, Taeuk Kang, Sangji Moon, Wonseok Leee, Sung H. Han, and Seungmoon Choi

Spring Conference, The Acoustical Society of Korea · 2024

4. Real-Time, Selective, and Multimodal Haptic Feedback from Gameplay Sound

Gyeore Yun, Jungeun Lee, Dong-Geun Kim, Minjae Mun, Hong Z. Tan, and Seungmoon Choi

IEEE World Haptics Conference · 2023

5. 음향-촉감 교차 모달 스펙트럼 매칭

Dong-Geun Kim, Jungeun Lee, Gyeore Yun, Hong Z. Tan, and Seungmoon Choi

Korean Institute of Next Generation Computing · 2023

Funded projects

Tactile Korean System for the Visually and Hearing Impaired (PM)

IITP | Jan. 2025 - Present

Conducting research on tactile delivery of Korean language using vibrotactile arrays to support communication for people with sensory disabilities.

Mid-air Haptic Rendering for Digital Twin (PM)

IITP | Jan. 2023 - Dec. 2024

Leading the development of real-time mid-air haptic rendering technologies. Researched enhancing perceived intensity of weak stimuli using cognitive modulation.

Investigating the Luxuriousness of Haptic Warning

Hyundai Motors | Apr. 2023 - Apr. 2024

Explored user perception of luxurious haptic warning feedback on steering wheels.

Performance Metric for User Experience

Google | Sep. 2021 - Jul. 2023

Designed metrics to assess UX in digital environments. Improved UX through cross-modal matching of sound and touch.

Multimodal Haptic Tool for AR Sports

KOCCA | Jun. 2020 - Dec. 2022

Designed devices rendering realistic haptic sensations by combining vibration, shear force, and impact feedback.

Technical skills

Programming: Python, Unity (C#), MATLAB, C/C++

Hardware & research: Arduino / Embedded, Fusion 360 (CAD), 3D Printing, Signal Processing, User Study Design

Languages

Korean (Native) · English (Fluent) · TOEIC 920 / TEPS 379